

MUSTAFACODE

Lecture 06

Object Oriented Programming Chapter 06 Detailed Notes (Roman Urdu)

Object-Oriented Programming (CS304)

Course Code: **CS304**

Publisher: **mustafacode**

Compiled On: **May 29, 2026**

Lecture 6: Object Oriented Programming Chapter 06 Detailed Notes (Roman Urdu)

Object Oriented Programming – Chapter 06 Detailed Notes (Roman Urdu)

06.1 Class Compatibility

Definition

Aik class dusri class ke sath **behaviorally compatible** tab hoti hai jab woh dusri class ke tamam operations ko support kare.

Aisi class ko **Subtype** kaha jata hai.

Matlab:

Agar derived class base class ke tamam kaam kar sakti hai to derived class, base class ka subtype hoti hai.

Important Point

Derived class ko base class ki jagah use kiya ja sakta hai.

Iska reason:

- Derived class base class ki properties aur operations inherit karti hai.
 - Is liye derived class base class ke tamam legal messages handle kar leti hai.
-

Real Life Example

Example 1: Vehicle

```
Vehicle
├─ Car
├─ Bike
└─ Bus
```

Agar Vehicle class mein operation ho:

```
start()
```

To Car, Bike aur Bus sab `start()` operation perform kar sakte hain.

Is liye:

- Car is subtype of Vehicle
- Bike is subtype of Vehicle
- Bus is subtype of Vehicle

06.2 Polymorphism

Definition

Polymorphism ka matlab hai:

"Same message but different behavior"

Ya:

Aik hi operation different objects par different tareeqe se kaam kare.

General Meaning

Real world mein bhi aik cheez ki multiple forms hoti hain.

Example

- Diamond
- Coal

Dono Carbon ki forms hain.

06.3 Polymorphism in OO Model

OO model mein different objects aik hi message ka different response dete hain.

Sender ko receiver ki exact class jaanne ki zarurat nahi hoti.

Sender sirf message send karta hai. Receiver apne hisaab se correct method execute karta hai.

Simple Programming Example

```
cat.speak()  
dog.speak()  
cow.speak()
```

Output

```
Cat   -> Meow  
Dog   -> Bark  
Cow   -> Moo
```

Yahan:

- Message same hai = `speak()`
- Behavior different hai

Isi ko polymorphism kehte hain.

Real Life Example

Remote Control Example

Aik remote ka button same hota hai:

```
Power ON
```

Lekin:

- TV ON ho jata hai
- AC ON ho jata hai
- Fan ON ho jata hai

Button same hai lekin behavior alag hai.

Yeh polymorphism hai.

Sender aur Receiver

Sender

Jo message bhejta hai.

Receiver

Jo message receive karta hai.

Example

```
shape.draw()
```

Yahan:

- `draw()` message hai
- Shape receiver hai

Agar object Circle hua to circle draw hoga. Agar object Triangle hua to triangle draw hoga.

Graphic Editor Example

Problem Statement

Aik graphic editor banana hai jo:

- Line draw kare
- Circle draw kare
- Triangle draw kare

User:

- Shape select kar sake
- Shape move kar sake
- Shape rotate kar sake

Shapes ko group bhi kiya ja sake.

Group aik single shape ki tarah behave kare.

Step 1 – Identify Classes

Classes kya hoti hain?

Classes objects ka blueprint hoti hain.

Problem statement mein nouns dhoonde jate hain.

Extracted Nouns

- Editor
- User
- Shape
- Line
- Circle
- Triangle
- Group

- Menu
- View

Step 2 – Remove Irrelevant Classes

Editor

Editor poore system ka naam hai. Is liye is ka object nahi banega.

User

User system ke bahar hai. Yeh external actor hai.

Final Classes

- Shape
 - Line
 - Circle
 - Triangle
 - Group
 - Menu
 - View
-

Step 3 – Identify Associations

Association

Objects ke darmiyan relationship ko association kehte hain.

Example

Group Relationship

Statement:

Individual shapes can be grouped together

Is se pata chala:

- Group contains Line
- Group contains Circle
- Group contains Triangle

Composition Relationship

Agar aik object dusre objects par strongly depend kare to usay composition kehte hain.

Example

Group ke andar:

- Shapes
- Other groups

ho sakte hain.

Aggregation Relationship

View Relationship

View:

- Lines show karta hai
- Circles show karta hai
- Triangles show karta hai
- Groups show karta hai

Is liye:

View HAS-A Shapes

Yeh aggregation hai.

Simple Association

Menu aur View

Menu View ko message bhejta hai.

```
Menu ---> View
```

Yeh One-Way Simple Association hai.

Step 4 – Identify Attributes

Attributes kya hote hain?

Object ki properties ko attributes kehte hain.

Shape Attributes

Line

- Color
 - Vertices
 - Length
-

Circle

- Color
 - Vertices
 - Radius
-

Triangle

- Color
- Vertices

- Angle
-

Shape

- Color
 - Vertices
-

Group

- noOfObjects
-

View

- noOfObjects
 - selected
-

Menu

- name
 - isOpen
-

Step 5 – Identify Operations

Operations kya hoti hain?

Objects ke actions ko operations kehte hain.

Common Operations

Shape

- Draw
- Select

- Move
 - Rotate
-

Line

- Draw
 - Select
 - Move
 - Rotate
-

Circle

- Draw
 - Select
 - Move
 - Rotate
-

Triangle

- Draw
 - Select
 - Move
 - Rotate
-

Group

- Draw
 - Select
 - Move
 - Rotate
-

Menu

- Open
 - Select
 - Move
 - Rotate
-

View Operations

- Add
 - Remove
 - Group
 - Show
 - Select
 - Move
 - Rotate
-

Real Life Example of Operations

Car Example

Attributes

- color
- model
- speed

Operations

- start()
 - stop()
 - accelerate()
 - brake()
-

Step 6 – Identify Inheritance

Inheritance kya hoti hai?

Jab aik class dusri class ki properties aur operations inherit kare.

Keywords

Inheritance identify karne ke liye:

- such as
- for example
- is a kind of

jaise words dhoonde jate hain.

Example

Shapes such as Line, Circle and Triangle

Matlab:

- Line is a Shape
 - Circle is a Shape
 - Triangle is a Shape
-

Inheritance Hierarchy

```
Shape
├─ Line
├─ Circle
├─ Triangle
└─ Group
```

Group Inherits from Shape

Statement:

Group behaves like a single shape

Is liye:

Group IS-A Shape

Refining the Object Model

Inheritance apply karne ke baad model ko improve kiya jata hai.

Shared Associations

View

View har tarah ki shape contain karta hai.

Group

Group har tarah ki shape contain karta hai.

Shared Attributes

Shape ki attributes:

- color
- vertices

Automatically inherit honghi:

- Line
- Circle

- Triangle
 - Group
-

Shared Operations

Shape ki operations:

- select()
- move()
- rotate()

Automatically inherit hongi.

Overriding

Draw Operation

Sab classes mein draw operation hai.

Lekin:

- Circle alag tareeqe se draw hota hai
- Triangle alag tareeqe se draw hota hai
- Line alag tareeqe se draw hoti hai

Is liye draw method override hota hai.

Real Life Example of Overriding

Animal Example

```
class Animal {  
    sound()  
}  
  
class Cat extends Animal {  
    sound() => Meow  
}  
  
class Dog extends Animal {  
    sound() => Bark  
}
```

Yahan:

- Method same hai = `sound()`
- Behavior different hai

Yeh overriding + polymorphism hai.

Important Definitions

Class

Objects ka blueprint.

Object

Class ka instance.

Attribute

Object ki property.

Operation

Object ka action/function.

Association

Objects ke darmiyan relationship.

Aggregation

Weak HAS-A relationship.

Composition

Strong HAS-A relationship.

Inheritance

IS-A relationship.

Polymorphism

Same message, different behavior.

Overriding

Child class ka parent method ko apne tareeqe se implement karna.

Short Summary

Class Compatibility

Derived class base class ki jagah use ho sakti hai.

Polymorphism

Same function different objects par different result deta hai.

Association

Objects ke darmiyan relation.

Attributes

Object ki properties.

Operations

Object ke actions.

Inheritance

Child class parent class ki cheezein inherit karti hai.

Overriding

Child class apna own implementation deti hai.

MUSTAFA CO